CS 1470 · DEEP LEARNING · FINAL PROJECT

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Multi-Agent Reinforcement Learning in Pommerman: *Comparing PPO and Dreamer-Style World Models*

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SECTION 01

Introduction

Multi-agent environments are harder than standard single-agent RL because the world changes both through physics and through other agents' decisions. A policy must learn what to do, while a model-based agent must also learn dynamics that include opponents.

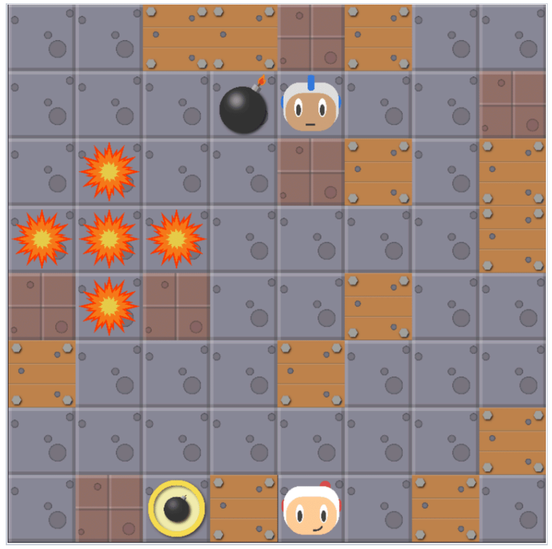
We study four-player free-for-all Pommerman by comparing a model-free PPO baseline with Dreamer-style world-model agents. The final Oscar runs changed the original local story: PPO was a useful early sanity check, but Shared Dreamer with horizon **h=3** became the strongest comparable FFA result.

SECTION 02

Game Environment

Pommerman is a multi-agent grid-world environment inspired by Bomberman, introduced by Resnick et al. in *Pommerman: A Multi-Agent Playground*. Agents move through a maze, place bombs, collect power-ups, and try to survive while eliminating opponents.

It is a useful testbed for multi-agent reinforcement learning because success requires both short-term reactions and longer-term planning: agents must avoid explosions, navigate around walls, anticipate opponent behavior, and decide when to act aggressively or defensively. Its combination of discrete actions, spatial observations, sparse rewards, and direct agent interaction makes it more challenging than many standard single-agent RL benchmarks while still being compact enough for controlled experimentation.

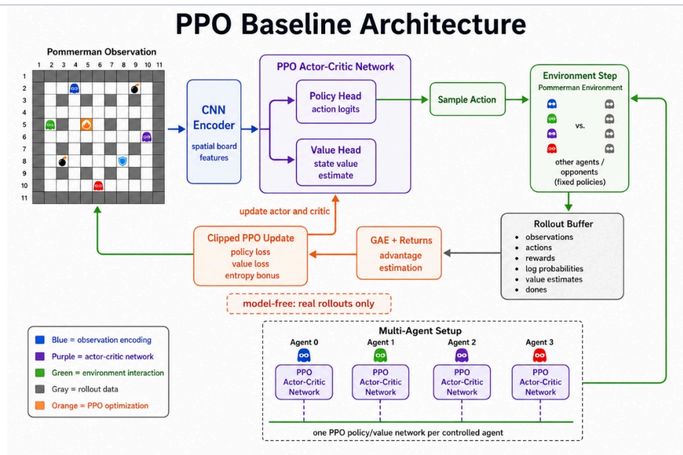


[FIGURE 1] The Pommerman environment: agents navigate a grid of destructible wooden walls and indestructible rigid walls, placing bombs that explode after a short fuse to clear walls and eliminate opponents.

SECTION 03

Architecture

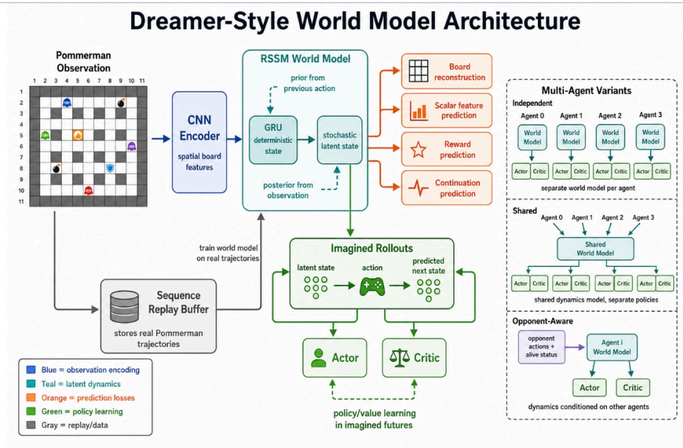
Our model suite compares a model-free PPO baseline against several Dreamer-style world model agents. The PPO agents use a convolutional encoder over Pommerman board observations, followed by separate policy and value heads trained with clipped policy updates and generalized advantage estimation. The Dreamer-style agents instead learn a recurrent state-space world model: observations are encoded with a CNN, passed through a GRU-based latent dynamics model, and decoded to predict future board states, rewards, and episode continuation. Policies and critics are then trained on imagined rollouts from this learned latent model.



PPO Baseline

MODEL-FREE

CNN encoder → actor-critic heads → clipped PPO update with GAE. One policy/value network per controlled agent; real rollouts only.



Dreamer-Style World Model

MODEL-BASED

CNN → RSSM (GRU + stochastic latent) predicts board, reward, and continuation. Actor/critic train on imagined rollouts.

Multi-Agent Variants

To study multi-agent structure, we implemented three world model strategies:

VARIANT A

Independent

Each player has its own world model and policy.

VARIANT B

Shared

Agents share a common world model but keep separate actor-critic heads.

VARIANT C

Opponent-Aware

Each agent's dynamics model is conditioned on other agents' recent actions and alive status.

SECTION 05

Challenges

The hardest part was making the whole multi-agent RL pipeline trustworthy enough to interpret. Pommerman forced us to handle environment wrapping, observation encoding, reward shaping, recurrent replay, action validity, checkpointing, logging, evaluation, and visualization.

Reward shaping helped agents move beyond passive survival, but it also made interpretation harder: PPO could improve dense reward without reliably winning. For Dreamer, decreasing world-model loss did not automatically produce good control. We therefore report shaped reward and terminal win rate separately, and track both best and final checkpoints because several Oscar runs spiked and then collapsed.

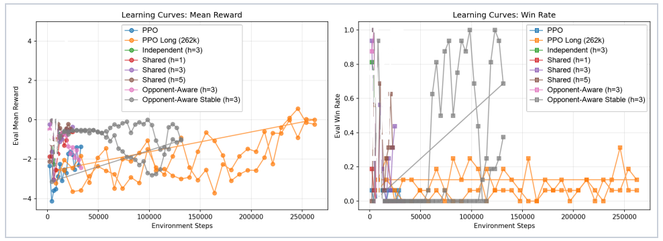
SECTION 06

Results

The 2048-step local study initially favored PPO, which reached **1.14** mean reward and a **50%** win rate while early Dreamer variants got zero wins. The larger Oscar FFA matrix changed the conclusion: short PPO ended at **0%** final win rate and PPO Long at **6.25%**, while Shared Dreamer **h=3** finished at **43.75%** and reached a **93.75%** best logged win rate.

43.75%	93.75%	6.25%
SHARED H=3 FINAL WIN	SHARED H=3 BEST WIN	PPO LONG FINAL WIN

OSCAR FFA RUN	FINAL WIN	BEST WIN
PPO	0%	6.25%
PPO Long	6.25%	31.25%
Independent Dreamer h=3	0%	81.25%
Shared Dreamer h=1	0%	18.75%
Shared Dreamer h=3	43.75%	93.75%
Shared Dreamer h=5	31.25%	68.75%
Opponent-Aware h=3	0%	100%
Opponent-Aware Stable h=3	37.5%	100%



[FIGURE 2] Oscar reward and win-rate curves. Win rate is noisy, but Shared Dreamer h=3 and stabilized opponent-aware Dreamer reach much stronger FFA win rates than PPO at their best checkpoints.

SECTION 07

Discussion & Future Work

The shared Dreamer results are the strongest positive finding. They suggest that Pommerman contains common physical dynamics that can be learned more efficiently by a shared model than by separate independent models. The horizon ablation also matters: h=1 was too short, h=3 performed best, and h=5 remained strong but slightly weaker.

Opponent-aware models were promising but high variance: they reached strong peaks, then often lost performance by the final checkpoint. Future work should prioritize **more seeds**, **larger evaluation batches**, **better checkpoint selection**, **stronger PPO masking and schedules**, and richer opponent modeling or curriculum learning.

REFERENCES [1]Resnick et al. (2018). *Pommerman: A Multi-Agent Playground.* [2]Hafner et al. (2020). *Dream to Control: Learning Behaviors by Latent Imagination.*